LIANG, "ERIC" FUHENG (梁赋珩)

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Astronomisches Rechen-Institut, Zentrum für Astronomie der Universität Heidelberg

EMPLOYMENT

Postdoctoral researcher, ARI, Heidelberg University Nov. 2024 – est. Oct. 2027
Advised by Kathryn Kreckel, working on galaxy evolution with precise and accurate metal abundances in the interstellar medium as part of PHANGS and SDSS-V/LVM collaborations.

Other fellowships offered: ASIAA Distinguished Fellowship (Taipei), Dunlap Fellowship (Toronto), and Faculty of Arts & Science Postdoctoral Fellowship (Toronto).

EDUCATION

DPhil, Dept. of Physics, University of Oxford Oct. 2020 – Oct. 2024
Thesis supervised by Martin Bureau, *Unravelling giant molecular clouds.*
Studentship at European Southern Observatory (Garching), Aug. 2023 – Aug. 2024, working with Ashley Barnes and Eric Emsellem.

Master, Dept. of Physics, Tsinghua University (Beijing) Sept. 2017 – Aug. 2020
Major in astrophysics, GPA: 3.9/4.0 (38 credits).
Thesis supervised by Cheng Li, *Studying Wolf-Rayet galaxies and the initial mass function with integral-field-unit survey SDSS-IV/MaNGA.*

Bachelor, Dept. of Physics, Tsinghua University (Beijing) Aug. 2013 – Jul. 2017
Primary major in physics, GPA: 3.6/4.0 (201 credits).
Thesis supervised by Cheng Li, *A new catalog of Wolf-Rayet galaxies from SDSS-IV/MaNGA.*
Secondary major in economics (47 credits).

PUBLICATIONS

First-author papers: 4; co-author papers: 40; h-index: 14.
SciX/ADS Library; ORCID 0000-0003-2496-1247.

4. **Liang, Fu-Heng**; Bureau, Martin; Liu, Lijie, et al., 2026, MNRAS, 547, stag221, *WISDOM Project - XXVII. Giant molecular clouds of the lenticular galaxy NGC 1387: similarities with spiral galaxy clouds*
3. **Liang, Fu-Heng**; Smith, Mark D.; Bureau, Martin, et al., 2024, MNRAS, 527, 9343, *WISDOM project - XVIII. Molecular gas distributions and kinematics of three megamaser galaxies*
2. **Liang, Fu-Heng**; Li, Cheng; Li, Niu, et al., 2021, ApJ, 923, 120, *Wolf-Rayet Galaxies in SDSS-IV MaNGA. II. Metallicity Dependence of the High-mass Slope of the Stellar Initial Mass Function*
1. **Liang, Fu-Heng**; Li, Cheng; Li, Niu, et al., 2020, ApJ, 896, 121, *Wolf-Rayet Galaxies in SDSS-IV MaNGA. I. Catalog Construction and Sample Properties*

40. Méndez-Delgado, J. Eduardo; Morisset, Christophe; Henney, William J., et al., 2026, arXiv: 2607.07973, *There is no single density: star-forming regions and galaxies hold more dense ionized gas than long assumed*
39. SDSS Collaboration; Aghakhanloo, Mojgan; Aird, James, et al., 2026, ApJS, 285, 9, *The Nineteenth Data Release of the Sloan Digital Sky Survey*

38. Kreckel, K.; Egorov, O. V.; Drory, N., et al., 2026, A&A, 710, A205, *SDSS-V LVM: Verifying what, and where, the "Galactic center lobe" is*

37. Dominiak, Pandora; Bureau, Martin; **Liang, Fu-Heng**, et al., 2026, MNRAS, 548, stag546, *WISDOM project - XXVIII. Molecular gas measurement of the supermassive black hole mass of the galaxy NGC 1387*

36. Hassani, Hamid; Rosolowsky, Erik; Leroy, Adam K., et al., 2026, ApJS, 284, 3, *The Hidden Life of Stars: Embedded Beginnings to Asymptotic Giant Branch Endings in the PHANGS-JWST Sample. I. Catalog of Mid-infrared Sources*

35. Scheuermann, Fabian; Kreckel, Kathryn; Teh, Jia Wei, et al., 2026, A&A, 708, A202, *Stellar associations powering H II regions: II. Escape fraction of ionising photons*

34. Hsieh, Pei-Ying; Walker, Daniel L.; Ginsburg, Adam, et al., 2026, arXiv:2603.00863, *ALMA Central molecular zone Exploration Survey (ACES) V: CS(2–1), SO(2_3–1_2), CH_3CHO($5_{1,4}$–$4_{1,3}$), HC_3N(11–10), and H40α lines data*

33. Lu, Xing; Walker, Daniel L.; Ginsburg, Adam, et al., 2026, arXiv:2602.20445, *ALMA Central Molecular Zone Exploration Survey (ACES)-IV. Data of the two intermediate-width spectral windows*

32. Barnes, A. T.; Chandar, R.; Kreckel, K., et al., 2026, A&A, 706, A95, *The PHANGS-MUSE/HST-Hα nebulae catalogue: Parsec-scale resolved structure, physical conditions, and stellar associations across nearby galaxies*

31. Sattler, Natascha; Méndez-Delgado, J. Eduardo; Kreckel, Kathryn, et al., 2026, A&A, 706, A81, *SDSS-V LVM: Resolving physical conditions in the Trifid Nebula*

30. Walker, Daniel L.; Ginsburg, Adam; Barnes, Ashley T., et al., 2026, arXiv:2602.20276, *ALMA Central molecular zone Exploration Survey (ACES) III: Molecular line data reduction and HNCO and HCO^+ data*

29. Ginsburg, Adam; Walker, Daniel L.; Barnes, Ashley T., et al., 2026, arXiv:2602.20240, *ALMA Central Molecular Zone Exploration Survey (ACES) II: 3mm continuum images*

28. Longmore, Steven N.; Bally, John; Barnes, Ashley T., et al., 2026, arXiv:2602.20340, *ALMA Central Molecular Zone Exploration Survey (ACES) I: Overview*

27. González-Torà, G.; Sander, A. A. C.; Egorova, E., et al., 2025, A&A, 703, L11, *SDSS-V LVM: Detectability of Wolf-Rayet stars and their He II ionizing flux in low-metallicity environments: I. The weak-lined, early-type WN3 stars in the SMC*

26. Kreckel, K.; Rickards Vaught, R. J.; Egorov, O. V., et al., 2025, A&A, 703, A42, *Temperature-based radial metallicity gradients in nearby galaxies*

25. Oakes, Elias K.; Faesi, Christopher M.; Rosolowsky, Erik, et al., 2025, ApJ, 993, 193, *The Hierarchical Dynamical State of Molecular Gas from 3 to 300 pc in NGC 253*

24. Liu, Lijie; Shu, Fanglin; Bureau, Martin, et al., 2025, MNRAS, 541, 3081, *WISDOM Project - XXIV. Giant molecular clouds of the spiral galaxy NGC 5064: high fraction of retrograde rotation*

23. den Brok, Jakob; Oakes, Elias K.; Leroy, Adam K., et al., 2025, ApJ, 988, 162, *Constraining Resolved Extragalactic R_{21} Variation with Well-calibrated ALMA Observations*

22. Lu, Anan; Haggard, Daryl; Bureau, Martin, et al., 2025, MNRAS, 540, 71, *WISDOM project - XXIII. Star formation efficiencies of eight early-type galaxies and bulges observed with SITELLE and ALMA*

21. Williams, Thomas G.; Belfiore, Francesco; Bureau, Martin, et al., 2025, MNRAS, 538, 3219, *The resolved star-formation efficiency of early-type galaxies*

20. Gao, Yang; Wang, Enci; Tan, Qing-Hua, et al., 2025, ApJ, 979, 105, *The First Exploration of the Correlations Between WISE 12 μm and CO Emission in Early-type Galaxies*

19. Neumann, Lukas; Jiménez-Donaire, María J.; Leroy, Adam K., et al., 2025, A&A, 693, L13, *Dense gas scaling relations at kiloparsec scales across nearby galaxies with the ALMA ALMOND and IRAM 30 m EMPIRE surveys*

18. Nonhebel, M.; Barnes, A. T.; Immer, K., et al., 2024, A&A, 691, A70, *Disruption of a massive molecular cloud by a supernova in the Galactic Centre: Initial results from the ACES project*

17. Lu, Anan; Haggard, Daryl; Bureau, Martin, et al., 2024, MNRAS, 531, 3888, *WISDOM project XX. - Strong shear tearing molecular clouds apart in NGC 524*

16. Verwilghen, Pierrick; Emsellem, Eric; Renaud, Florent, et al., 2024, A&A, 687, A53, *Simulating nearby disc galaxies on the main star formation sequence. I. Bar formation and the building of the central gas reservoir*

15. Choi, Woorak; Bureau, Martin; Liu, Lijie, et al., 2024, MNRAS, 531, 4045, *WISDOM Project - XXI. Giant molecular clouds in the central region of the barred spiral galaxy NGC 613: a steep size-linewidth relation*

14. Zhang, Hengyue; Bureau, Martin; Smith, Mark D., et al., 2024, MNRAS, 530, 3240, *WISDOM Project - XIX. Figures of merit for supermassive black hole mass measurements using molecular gas and/or megamaser kinematics*

13. Ruffa, Ilaria; Davis, Timothy A.; Elford, Jacob S., et al., 2024, MNRAS, 528, L76, *A fundamental plane of black hole accretion at millimetre wavelengths*

12. Elford, Jacob S.; Davis, Timothy A.; Ruffa, Ilaria, et al., 2024, MNRAS, 528, 319, *WISDOM Project - XVI. The link between circumnuclear molecular gas reservoirs and active galactic nucleus fuelling*

11. Williams, Thomas G.; Bureau, Martin; Davis, Timothy A., et al., 2023, MNRAS, 525, 4270, *WISDOM Project - XVII. Beam-by-beam properties of the molecular gas in early-type galaxies*

10. Ruffa, Ilaria; Davis, Timothy A.; Cappellari, Michele, et al., 2023, MNRAS, 522, 6170, *WISDOM project - XIV. SMBH mass in the early-type galaxies NGC 0612, NGC 1574, and NGC 4261 from CO dynamical modelling*

9. Choi, Woorak; Liu, Lijie; Bureau, Martin, et al., 2023, MNRAS, 522, 4078, *WISDOM Project - XV. Giant molecular clouds in the central region of the barred spiral galaxy NGC 5806*

8. Liu, Lijie; Bureau, Martin; Li, Guang-Xing, et al., 2022, MNRAS, 517, 632, *WISDOM Project - XII. Clump properties and turbulence regulated by clump-clump collisions in the dwarf galaxy NGC 404*

7. Lu, Anan; Boyce, Hope; Haggard, Daryl, et al., 2022, MNRAS, 514, 5035, *WISDOM project - XI. Star formation efficiency in the bulge of the AGN-host Galaxy NGC 3169 with SITELLE and ALMA*

6. Davis, Timothy A.; Gensior, Jindra; Bureau, Martin, et al., 2022, MNRAS, 512, 1522, *WISDOM Project - X. The morphology of the molecular ISM in galaxy centres and its dependence on galaxy structure*

5. Abdurro'uf; Accetta, Katherine; Aerts, Conny, et al., 2022, ApJS, 259, 35, *The Seventeenth Data Release of the Sloan Digital Sky Surveys: Complete Release of MaNGA, MaStar, and APOGEE-2 Data*

4. Li, Niu; Li, Cheng; Mo, Houjun, et al., 2021, ApJ, 917, 72, *Estimating Dust Attenuation From Galactic Spectra. II. Stellar and Gas Attenuation in Star-forming and Diffuse Ionized Gas Regions in MaNGA*

3. Xi, Hongwei; Staveley-Smith, Lister; For, Bi-Qing, et al., 2021, MNRAS, 501, 4550, *The Arecibo Ultra-Deep Survey*
2. Albareti, Franco D.; Allende Prieto, Carlos; Almeida, Andres, et al., 2017, ApJS, 233, 25, *The 13th Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the SDSS-IV Survey Mapping Nearby Galaxies at Apache Point Observatory*
1. Blanton, Michael R.; Bershady, Matthew A.; Abolfathi, Bela, et al., 2017, AJ, 154, 28, *Sloan Digital Sky Survey IV: Mapping the Milky Way, Nearby Galaxies, and the Distant Universe*

AWARDS AND GRANTS

2023 – 2024, European Southern Observatory studentship (Garching).

2023 – 2024, Scatcherd European Scholarship, University of Oxford.

2019, Tsinghua–AMD Space Science and Technology Awards, first prize.

2017, Chi-Sun Yeh Prize (叶企孙奖), the highest honour for undergraduate students at the Department of Physics, Tsinghua University.

2016, Scholarship of National Astronomical Observatories, Chinese Academy of Sciences.

2015, Undergraduate Exchange Programme, China Scholarship Council.

2014 – 2017, Xuetang Talents Programme, Tsinghua University.

CONFERENCES AND TALKS

Sept. 2026, Annual Meeting of the Astronomische Gesellschaft (German Astronomical Society). Contributed talks "Using Wolf-Rayet regions in nearby galaxies to study star formation and stellar feedback" and "WISDOM in ALMA2040: galaxy centre physics at high resolution, from supermassive black holes to molecular clouds".

Jan. 2026, ARI colloquium "Cold molecular gas (cloud) properties in nearby galaxies" at Heidelberg University.

Jul. 2025, ESO Workshop "Galactic ecosystems", ESO (Garching). Contributed talk "Giant molecular clouds in passive lenticular galaxy NGC1387" and poster "Using Wolf-Rayet regions as a unique laboratory to study star formation and stellar feedback".

Apr. 2025, invited talk "Resolving interstellar medium — early results from Local Volume Mapper" at National Astronomical Observatories of the Chinese Academy of Sciences.

Jan. 2025, colloquium "Cold molecular gas (cloud) properties in nearby galaxies" at Academia Sinica Institute of Astronomy and Astrophysics and at The Chinese University of Hong Kong.

Sept. 2024, Wine & Cheese Seminar, ESO (Garching). Invited talk "Interplay of cold molecular gas, dynamic shear, and magnetic field in Milky Way centre".

Feb. 2024, Galaxy Evolution Coffee Seminar, MPA/MPE/ESO (Garching). Invited talk "Giant molecular clouds in passive lenticular galaxy NGC1387 and conundrum of star formation".

Jan. 2024, IR/mm Group Seminar, Max Planck Institute for Extraterrestrial Physics (Garching). Invited talk "Cold molecular gas (cloud) properties across different types of galaxies".

Nov. 2023, Galaxy Coffee Seminar, Max Planck Institute for Astronomy (Heidelberg). Invited talk "Cold molecular gas (cloud) properties across different types of galaxies".

Jul. 2023, National Astronomy Meeting, Cardiff University. Co-organiser of parallel session "Gas in and around galaxies".

Jun. 2023, invited talks "Cold molecular gas (cloud) properties across different types of galaxies" at Tsinghua University, Kavli Institute for Astronomy and Astrophysics (PKU), Shanghai Jiao Tong University, Nanjing University, National Astronomical Observatories & Shanghai Astronomical Observatory of the Chinese Academy of Sciences.

Sept. 2022, 4th Philip Wetton Workshop, University of Oxford. SOC/LOC member.

May 2022, Ringberg Seminar Series. Invited talk "Giant molecular clouds in passive lenticular galaxy NGC 1387".

Jun. 2021, Ringberg Seminar Series. Invited talk "Constraining the initial mass function with Wolf-Rayet galaxies in MaNGA".

Jan. 2021, Galaxy Evolution Seminar, University of Oxford. Invited talk "Metallicity dependence of the stellar initial mass function revealed by Wolf-Rayet galaxies".

Apr. 2019, MaNGA Collaboration Meeting, University of Oxford. Contributed talk "Wolf-Rayet galaxies in MaNGA".

Jun. 2018, SDSS-IV collaboration meeting, Sejong University. Lightning talk and poster "Wolf-Rayet galaxies in MaNGA".

VISITS

Jan. 2025, ASIAA (Taipei); host: Lihwai Lin.

Nov. 2023, Max Planck Institute for Astronomy (Heidelberg); host: Eva Schinnerer.

Aug. 2022, Yonsei University (Seoul); host: Aeree Chung.

Jul. – Aug. 2016, University of Kentucky (Lexington), working on Wolf-Rayet galaxy catalogue from MaNGA; host: Renbin Yan.

Apr. 2016, ESO (Garching), working on Arecibo Ultra Deep Survey; host: Martin Zwaan.

Sept. 2015 – Dec. 2015, UBC (Vancouver), undergraduate exchange programme.

Jul. – Aug. 2015, ICRAR (Perth), Arecibo Ultra-Deep Survey; host: Lister Staveley-Smith.

OBSERVATION RELATED

Accepted PI ALMA observing proposals (three): 2022.1.01173.S, 2023.1.01084.S, 2024.1.01104.S.

Accepted co-I observing proposals: one with *JWST*, one joint VLA-*JWST*, one with VLT/MUSE, two with CFHT/SITELLE, nine with ALMA, one with IRAM, one with SMA, one with SDSS-V open-tile, and two with VLA.

2023 – 2024, Panel Meeting Scientific Assistant of ESO Observing Programmes Committee.

3rd Feb. 2016, guest observer, JCMT, East Asian Observatory (Hawaii), host: Iain Coulson.

SERVICE, TEACHING, AND OUTREACH

2026, supervising Master's student S. Aji at Heidelberg University on "Galactic Wolf-Rayet nebula around WR128".

2025 – present, organising PHANGS-Heidelberg monthly tag-up meeting.

2023 – 2024, organising committee, weekly joint Galaxy Evolution Coffee at ESO/MPE/MPA.

2021 – 2024, organising WISDOM collaboration's weekly GMC/ISM meeting (hybrid).

2023, mentoring Master's student M. Maksymowicz-Maciata at Oxford on "Comparing circumnuclear regions of (ultra-)luminous infrared galaxies with CO and CI fluxes", first-class.

2021 – 2023, lab demonstrator of 3rd-yr undergraduate physics, Astrophysics Lab, Department of Physics, University of Oxford; promoted "Senior Demonstrator" in 2022.

2020 – 2023, homework marker of 1st-yr undergraduate physics, Wadham College, Oxford.

2021 – 2023, co-host of the "Academic Profile Development" workshop as part of the Oxford Prospects Programmes, Regent's Park College, University of Oxford.

2021, panel discussion member of Stargazing Oxford @ Home, with Dr. Becky et al.

2019, organising committee, Galaxy and Cosmology Seminar, Tsinghua University.

2019, translator (with others) of the Chinese version of outreach book *The Sun* (by Golub and Pasachoff).

2017 – 2018, stipendiary mentor of undergraduates, Department of Physics, Tsinghua University, evaluated "Distinction".